

Car Market Analysis Dashboard (Excel)

1. Title Page

Project Title: Car Market Analysis Dashboard

Name: Priyanshi Yadav

Submission Date: 14-06-2026

2. Objective

The objective of this project is to analyze the used car dataset using Microsoft Excel. The project focuses on data cleaning, data analysis using formulas, creation of pivot tables, and designing an interactive dashboard. The goal is to understand pricing trends, fuel distribution, ownership patterns, and seller performance.

3. Dataset Description

The dataset used for this project is the CarDekho used car dataset. It contains information about different cars, including attributes such as car name, manufacturing year, selling price, kilometers driven, fuel type, seller type, transmission type, and ownership details.

4. Data Preparation (Phase 1)

The following steps were performed to clean and prepare the dataset:

- Removed duplicate records from the dataset
 - Checked and handled missing values
 - Converted the selling price column into numeric format
 - Created a new column **Car Age** using the formula:
Car Age = Current Year – Year
 - Created a new column **Price Category** using IF condition:
 - Budget (Less than 5 Lakhs)
 - Mid Range (5–10 Lakhs)
 - Premium (Above 10 Lakhs)
-

5. Data Analysis (Phase 2)

The dataset was analyzed using Excel formulas (without using any lookup functions):

- Total number of cars was calculated using COUNTA
- Average selling price was calculated using AVERAGE

- Most common fuel type was identified using COUNTIF
 - Number of Manual and Automatic cars was calculated
 - Total cars by Seller Type were counted
 - Average selling price by Fuel Type was calculated using AVERAGEIF
 - Number of cars older than 10 years was calculated
 - Percentage of First Owner cars was computed
-

6. Pivot Tables (Phase 3)

The following pivot tables were created for better analysis:

- Average Selling Price by Fuel Type
 - Count of Cars by Transmission
 - Year-wise Car Count
 - Seller Type vs Average Selling Price
-

7. Dashboard Creation (Phase 4)

An interactive dashboard was created in a separate sheet with the following features:

- KPI Cards displaying:
 - Total Cars (3578)
 - Average Selling Price
 - Most Used Fuel (Diesel)
 - Percentage of First Owner Cars
 - Charts used:
 - Bar Chart (Fuel vs Average Price)
 - Column Chart (Transmission Count)
 - Line Chart (Year-wise Trend)
 - Bar Chart (Seller Type vs Average Price)
 - Slicer added for interactivity (Fuel/Transmission)
 - Proper formatting and alignment were maintained for a professional appearance
-

8. Insights

The following insights were derived from the analysis:

1. Diesel cars are the most commonly used fuel type in the dataset.
 2. Manual transmission cars are significantly higher in number than automatic cars.
 3. The number of cars has increased after 2010, indicating growth in the car market.
 4. Dealer-listed cars have a higher average selling price compared to individual sellers.
-

9. Conclusion

This project helped in understanding how to work with real-world data using Excel. It improved skills in data cleaning, using formulas, creating pivot tables, and designing dashboards. The project also provided insights into trends and patterns in the used car market.

10. File Submission

The Excel file is submitted in .xlsx format with the required naming convention:

PriyanshiYadav_RollNumber_CarDashboard.xlsx
